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Find the first and second derivatives.

$$y = -5x^4 + 7$$

To find $\frac{dy}{dx}$, first use the Derivative Sum Rule, which states if u and v are differentiable functions of x , then their sum $u + v$ is differentiable at

every point where u and v are both differentiable. At such points, $\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$.

$$\frac{dy}{dx} = \frac{d}{dx}(-5x^4 + 7) = \frac{d}{dx}(-5x^4) + \frac{d}{dx}(7)$$

To find $\frac{d}{dx}(-5x^4)$, use the Constant Multiple Rule and the Power Rule for Positive Integers.

The Constant Multiple Rule states that if u is a differentiable function of x , and c is a constant, then $\frac{d}{dx}(cu) = c \frac{du}{dx}$.

The Power Rule for Positive Integers states that if n is a positive integer, then $\frac{d}{dx}(x^n) = nx^{n-1}$.

$$\begin{aligned} \frac{d}{dx}(-5x^4) &= -5 \frac{d}{dx}(x^4) \\ &= -5 \cdot 4x^3 \\ &= -20x^3 \end{aligned}$$

To find $\frac{d}{dx}(7)$, use the Derivative of a Constant Function Rule, which states if f has the constant value $f(x) = c$, then $\frac{df}{dx} = \frac{d}{dx}(c) = 0$.

$$\frac{d}{dx}(7) = 0$$

$$\text{Thus, } \frac{dy}{dx} = \frac{d}{dx}(-5x^4 + 7) = -20x^3.$$

Now find $\frac{d^2y}{dx^2}$. Note that $f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{dy'}{dx} = y''$.

To find $\frac{d}{dx}(-20x^3)$, again use the Constant Multiple Rule and the Power Rule for Positive Integers.

$$\frac{d}{dx}(-20x^3) = -60x^2$$

$$\text{Thus, } \frac{dy}{dx} = -20x^3 \text{ and } \frac{d^2y}{dx^2} = -60x^2.$$